Quantum Field Theory IV Week 1

Exercise 1.6

Linear sigma model with O(N)-symmetry

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi^i)^2 - \frac{1}{2}\mu^2(\phi^i)^2 - \frac{\lambda}{4}\left[(\phi^i)^2\right]^2 + \frac{1}{2}\delta_Z(\partial_\mu\phi^i)^2 - \frac{1}{2}\delta_m(\phi^i)^2 - \frac{1}{4}\delta_\lambda\left[(\phi^i)^2\right]^2$$

Now we decompose the field as $\phi^i = \phi^i_{cl} + \eta^i$

it is most convinient to do this such that

$$\phi^i = \begin{pmatrix} \eta^1 \\ \eta^{N-1} \\ \eta^N + \phi_{cl} \end{pmatrix}$$ so that ϕ^i_{cl} points to the Nth direction

We take the classical field ϕ_{cl} to be a constant classical field since the vacuum state is expected to be translation invariant $\Rightarrow \partial_\mu\phi_{cl} = 0$

$\Rightarrow$ Now Lagrangian is:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\eta^i)^2 - \frac{1}{2}\mu^2(\phi^i_{cl}+\eta^i)^2 - \frac{\lambda}{4}\left[(\phi^i_{cl}+\eta^i)^2\right]^2 + \frac{1}{2}\delta_Z(\partial_\mu\eta^i)^2 - \frac{1}{2}\delta_m(\phi^i+\eta^i)^2 - \frac{1}{4}\delta_\lambda\left[(\phi^i)^2\right]^2$$

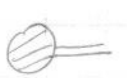
We need not care for the counter terms for this exercise so let's drop em and only examine the non counter terms

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\eta^i)^2 - \frac{1}{2}\mu^2(\phi^i_{cl})^2 - \frac{1}{2}\mu^2(2\phi^i_{cl}\eta^i) - \frac{1}{2}\mu^2(\eta^i)^2 - \frac{\lambda}{4}\left[(\phi^i_{cl})^2 + 2(\phi^i_{cl}\eta^i) + (\eta^i)^2\right]^2$$

$$= \frac{1}{2}(\partial_\mu\eta^i)^2 - \frac{1}{2}\mu^2(\eta^i)^2 - \frac{1}{2}\mu^2(\phi^i_{cl})^2 - \frac{\lambda}{4}\left[(\phi^i_{cl})^2\right]^2 - \left(\mu^2 - \lambda(\phi^i_{cl})^2\right)(\phi^i_{cl}\eta^i)$$

$$- \frac{\lambda}{2}\left[(\phi^i_{cl})^2(\eta^i)^2 + 2(\phi^i_{cl}\eta^i)^2\right] + \text{terms of order } (\eta^i)^3 \text{ or higher}$$

Since we want to work in scenario where the tadpole contribution to $\langle\eta\rangle$ is zero we work with the condition

 $+ \delta J($  $) = 0$, so we fix δJ

(The variation/fluctuation of the source), which means we can ignore the terms linear in η^i

Now let's look at the expansion of the action to get how $\frac{\delta^2 S}{\delta\phi^i\delta\phi^j}$ looks:

$$\int d^4x\left(\mathcal{L} + J\phi^i\right) = \int d^4x\, \mathcal{L}[\phi_{cl}] + J\phi_{cl} + \int d^4x\, \eta^i(x)\left(\frac{\delta S}{\delta\phi^i}\Big|_{\phi_{cl}} + J_i\right)$$

$$+ \frac{1}{2}\int d^4x\int d^4y\, \eta^i(x)\,\eta^j(y)\,\frac{\delta^2 S}{\delta\phi^i\delta\phi^j}\Big|_{\phi=\phi_{cl}} + \dots$$ as per the Lecturenotes

We get the $\frac{\delta^2 S}{\delta\phi^i \delta\phi^j}$ term by matching the terms of order $(\eta^i)^2$

In the Lagrangian these terms are : $-\frac{1}{2}\mu^2(\eta^i)^2$, $-\frac{\lambda}{2}\left[(\phi^i_{cl})^2(\eta^i)^2 + 2(\phi^i_{cl}\eta^i)^2\right]$

and $\frac{1}{2}(\partial_\mu \eta^i)^2 = -\frac{1}{2}\eta^i\partial^2\eta^i$ by integration by part where the surface term vanishes

The $-\frac{1}{2}\mu^2(\eta^i)^2$ and $-\frac{1}{2}\eta^i\partial^2\eta^i$ and $-\frac{\lambda}{2}(\phi^k_{cl})^2(\eta^i)^2$ contribute a δ^{ij}

since there is $\eta^i(x)\eta^i(y)$ term that is proportional to $\frac{\delta^2 S}{\delta\phi^i\delta\phi^i}$

the term $-\lambda(\phi^i_{cl}\eta^i)^2$ doesn't contribute δ^{ij} since we can write it as

$$-\lambda(\phi^i_{cl}\eta^i)^2 = -\lambda(\phi^i_{cl}\eta^i)(\phi^j_{cl}\eta^j)$$ since the sum is

between indices of ϕ_{cl} and η^i .

the $\int d^4y$ integral just makes sure that $y = x$ since $\frac{\delta^2 S}{\delta\phi^i(x)\delta\phi^j(y)}$

will produce a $\delta(x-y)$ term so that $\int d^4y \frac{\delta^2 S}{\delta\phi^i(x)\delta\phi^j(y)}\eta^i(x)\eta^j(y) \Rightarrow \frac{\delta^2 S}{\delta\phi^i\delta\phi^j}\eta^i\eta^j$

x-dependence only

This is because $\frac{\delta}{\delta J(x)} J(y) = \delta^{(4)}(x-y)$

and S is purely the function of $\phi(x)$. Here I am assuming this S from lecture notes is same as $\mathcal{L}_1$ from Peskin (11.58) since the expansion is identical .

From all of these above we now write

We wrote this summation index as k to avoid confusion with the free indices i and j

$$\left.\frac{\delta^2 S}{\delta\phi^i\delta\phi^j}\right|_{\phi=\phi_{cl}} = -\partial^2\delta^{ij} - \mu^2\delta^{ij} - \lambda\left[(\phi^k_{cl})^2\delta^{ij} + 2\phi^i_{cl}\phi^j_{cl}\right]$$

$\|\ \partial^2 = \Box$

Let's write $(M^2)^{ij}(\phi_{cl}) \equiv +\mu^2\delta^{ij} + \lambda\left[(\phi^k_{cl})^2\delta^{ij} + 2\phi^i_{cl}\phi^j_{cl}\right]$

which is $+\mu^2 + \lambda\phi^2_{cl}$ when acting on $\eta^1 \cdots \eta^{N-1}$

and $+\mu^2 + \lambda\phi^2_{cl} + 2\lambda\phi^2_{cl} = +\mu^2 + 3\lambda\phi^2_{cl}$ when acting on η^N

$$\Rightarrow \frac{\delta^2 S}{\delta\phi^i\delta\phi^j}[\phi_{cl}] = -\Box\delta^{ij} - (M^2)^{ij}(\phi_{cl})$$ $\Box$

Exercise 1.7

Compute the one-loop effective potential of linear sigma model.

Let's strart from the partition function, which after the expansion in previous exercise looks like:

$$Z[J] = \exp\left(i\int d^4x\left[\mathcal{L}_R[\phi_{cl}] + J_R(x)\,\phi_{cl} + \Delta\mathcal{L}[\phi_{cl}] + \Delta J_{(x)}\phi_{cl}\right]\right)\int D\eta_i\, e^{i(\tilde{S}[\eta] + \Delta\tilde{S}[\eta])}$$

There are N - amount of measures since we have N distinct η-fields

Where $\mathcal{L}_R[\phi_{cl}]$ is the part of the Lagrangian that only depends on ϕ_{cl} with $\Delta\mathcal{L}_R[\phi_{cl}]$ being it's counter-term.

$$\tilde{S}[\eta] = \int d^4x\int d^4y\ \frac{1}{2}\eta^i(x)\left(\frac{\delta^2 S}{\delta\phi^i\delta\phi^j}[\phi_{cl}]\right)\eta^j(y) + \dots$$

with $\Delta\tilde{S}[\eta]$ being the counter terms.

From previous exercise we got $\frac{\delta^2 S}{\delta\phi^i\delta\phi^j}[\phi_{cl}] = -\Box\delta^{ij} - (M^2)^{ij}(\phi_{cl})$

$$\Rightarrow\quad \tilde{S}[\eta] = \int d^4x\int d^4y\ \frac{1}{2}\eta^i(x)\left[-\Box\delta^{ij} - (M^2)^{ij}(\phi_{cl})\right]\eta^j(y)$$

$(M^2)^{ij} = \lambda\left[(\phi^k)^2\delta^{ij} + 2\phi^i_{cl}\phi^j_{cl}\right]$
So it's 0 when $i\neq j$
since $\phi^i = (0,0,\dots,0,\phi^N_{cl})$
So we can write in terms of just $(M^2)^i$

$\Rightarrow \int D\eta_i \exp(i\tilde{S}[\eta])$ is Gaussian

In the 1-loop order the fields don't interact and we basically have N- amount of Gaussian integrals. Since fields don't interact this result is enough (no need for perturbative interactions)

$$\Rightarrow \int D\eta_i \exp(i\tilde{S}[\eta]) = \det^{-\frac{N-1}{2}}\left(\Box + \lambda\phi^2_{cl}\right)\cdot\det^{-\frac{1}{2}}\left(\Box + 3\lambda\phi^2_{cl}\right)$$

$= -\frac{\delta^2 S}{\delta\phi\delta\phi}$

$$= \exp\left[-\frac{1}{2}(N-1)\operatorname{tr}\ln\left(\Box + \lambda\phi^2_{cl}\right) - \frac{1}{2}\operatorname{tr}\ln\left(\Box + 3\lambda\phi^2_{cl}\right)\right]$$

Now we have $Z[J] = e^{iW[J]}$

$$W[J] = \int d^4x\left[\mathcal{L}_R[\phi_{cl}] + J_R(x)\phi_{cl} + \Delta\mathcal{L}[\phi_{cl}] + \Delta J(x)\,\phi_{cl}\right] + \frac{i}{2}(N-1)\operatorname{tr}\ln(\Box + \lambda\phi^2_{cl}) + \frac{i}{2}\operatorname{tr}\ln(\Box + 3\lambda\phi^2_{cl})$$

Now let's do the Lagandre transformation to get $\Gamma(\phi_{cl}) = -(VT)\,V_{eff}$

$$\Rightarrow\quad \Gamma(\phi_{cl}) = W[J] - \int dx\ J(x)\phi_{cl}\quad,\text{where } J(x) = J_R + \Delta J$$

$$\Rightarrow\quad \Gamma(\phi_{cl}) = \int d^4x\left[\mathcal{L}_R[\phi_{cl}] + \Delta\mathcal{L}[\phi_{cl}]\right] + \frac{i}{2}(N-1)\operatorname{tr}\ln\left[\Box + \lambda\phi^2_{cl}\right] + \frac{i}{2}\operatorname{tr}\ln\left[\Box + 3\lambda\phi^2_{cl}\right]$$

So we have

$$\Gamma(\phi_{cl}) = S[\phi_{cl}] + \Delta S[\phi_{cl}] + \underbrace{i\frac{(N-1)}{2} tr\ ln\left[\Box + \mu^2 + \lambda\phi_{cl}^2\right]}_{\text{this are } V_{eff} \text{ part given by } i=1,\dots,N-1 \text{ fields } \eta^i} + \underbrace{\frac{i}{2} tr\, ln\left[\Box + \mu^2 + 3\phi_{cl}^2\right]}_{V_{eff} \text{ part given by } \eta^N \text{ field}}$$

We can write the Logarith parts as V_{eff}^i, so a V_{eff} contribution from ith field η, such that

$$V_{eff}^i = -\frac{i}{2}\, tr\, ln\left[\Box + M_i^2\right]\frac{1}{VT}, \text{ where } M_i^2 = \begin{cases} \mu^2 + \lambda\phi_{cl}^2 & , i=1,\dots,N-1 \\ \mu^2 + 3\lambda\phi_{cl}^2 & , i=N \end{cases}$$

So that

$$\Gamma(\phi_{cl}) = \int d^4x \left(\mathcal{L}_R[\phi_{cl}] + \Delta\mathcal{L}[\phi_{cl}] - \sum_i V_{eff}^i\right)$$

Now we get the total V_{eff} by just $V_{eff} = -\frac{1}{VT}\Gamma(\phi_{cl})\Big|_{\phi_{cl}=const}$

$$\Rightarrow \quad V_{eff} = -\frac{1}{VT}\Gamma(\phi_{cl}) = -\mathcal{L}_R[\phi_{cl}] - \Delta\mathcal{L}[\phi_{cl}] + \sum_i V_{eff}^i$$

|| $\int d^4x\, \mathcal{L}[\phi_{cl}] = VT\mathcal{L}[\phi_{cl}]$ since $\phi_{cl} = const$

Now let's expand the V_{eff}^i

$$V_{eff}^i = -\frac{i}{2}\frac{1}{VT}\, tr\, ln\left[\Box + M_i^2\right]$$

$$V_{eff}^i = -\frac{i}{2}\int\frac{d^4k}{(2\pi)^4}\, ln(-k^2 + M_i^2)$$

|| We evaluate the trace as sum of eigenvalues k and move to continuum $\left(tr \to \sum_k\right)$

$$\sum_k \to VT\int\frac{d^4k}{(2\pi)^4}$$

Let's wick rotate to evaluate this in the Euclidean space using dimensional regularization

$$\Rightarrow \quad V_{eff}^i = \frac{1}{2}\int\frac{d^dk_E}{(2\pi)^d}\, ln\left(k_E^2 + M_i^2\right)$$

|| $ln(x) = -\frac{d}{d\alpha}\frac{1}{x^\alpha}\Big|_{\alpha=0}$

$$= -\frac{1}{2}\partial_\alpha\int\frac{d^dk_E}{(2\pi)^d}\frac{1}{(k_E^2 + M_i^2)^\alpha}\Bigg|_{\alpha=0}$$

|| let's now use the known solution to this integral from Peskin (7.85)

$$= -\frac{1}{2}\partial_\alpha\left(\frac{1}{(4\pi)^{d/2}}\frac{\Gamma(\alpha - d/2)}{\Gamma(\alpha)}\frac{1}{(M_i^2)^{\alpha - d/2}}\right)\Bigg|_{\alpha=0}$$

$$V_{eff}^i = -\frac{1}{2}\frac{1}{(4\pi)^{d/2}}\frac{\Gamma(-d/2)}{(M_i^2)^{-d/2}}$$

|| $\Gamma(\alpha) \to \frac{1}{\alpha}$ as $\alpha \to 0$

$\Rightarrow \frac{1}{\Gamma(\alpha)} \to \alpha$ and $\Gamma(\alpha - d/2) \to \Gamma(-d/2)$

$(M^2)^{\alpha - d/2} \to (M^2)^{-d/2}$

So only $(\partial_\alpha\, \alpha)\big|_{\alpha=0} = 1$ remains

$$\Rightarrow V_{eff} = -\mathcal{L}_R[\phi_{cl}] - \Delta\mathcal{L}[\phi_{cl}] + \sum_i V^i_{eff}$$

$$= -\mathcal{L}_R[\phi_{cl}] - \Delta\mathcal{L}[\phi_{cl}] + \sum_i \left(-\frac{1}{2}\right) \frac{\Gamma(-d/2)}{(4\pi)^{d/2}} (M_i^2)^{d/2}$$

One loop effective potential

$$\boxed{V_{eff} = -\mathcal{L}_R[\phi_{cl}] - \Delta\mathcal{L}[\phi_{cl}] - \frac{1}{2} \frac{\Gamma(-d/2)}{(4\pi)^{d/2}} \left[(N-1)(\mu^2 + \lambda\phi_{cl}^2)^{d/2} + (\mu^2 + 3\lambda\phi_{cl}^2)^{d/2} \right]}$$

Let's now expand the $\frac{\Gamma(-d/2)}{(4\pi)^{d/2}} (M_i^2)^{d/2}$ term to extract the divergencies to get the counterterms

The pole at $d=0$ is not of physical significance since it gives an infinite constant to Lagrangian but constant terms don't matter. We can write

$\Gamma(1-d/2) = \left(-\frac{d}{2}\right)\Gamma(-d/2)$, The pole at $d=2$ is quadratic in ϕ_{cl}

Now let's determine $\Delta\mathcal{L}[\phi_{cl}]$ to get the counter terms

$$\Delta\mathcal{L}[\phi_{cl}] = \frac{1}{2}\delta_Z \underbrace{(\partial_\mu \phi^i_{cl})^2}_{=0} - \frac{1}{2}\delta_\mu (\phi^i_{cl})^2 - \frac{\lambda}{4}\left[(\phi^i_{cl})^2\right]^2 \quad \text{from Peskin (11.14)}$$

$$= -\frac{1}{2}\delta_\mu \phi_{cl}^2 - \frac{\lambda}{4}\phi_{cl}^4$$

|| here we wrote the $\delta_M \equiv \delta_\mu$ as the counter term

$$\boxed{\Rightarrow V_{eff} = -\mathcal{L}[\phi_{cl}] + \frac{1}{2}\delta_\mu \phi_{cl}^2 + \frac{\delta_\lambda}{4}\phi_{cl}^4 - \frac{1}{2}\frac{\Gamma(-d/2)}{(4\pi)^{d/2}}\left[(N-1)(\mu^2+\lambda\phi_{cl}^2)^{d/2} + (\mu^2+3\lambda\phi_{cl}^2)^{d/2}\right]}$$

Now for $d=2$ the divergence in $\Gamma(-d/2) = \frac{\Gamma(1-d/2)}{(-d/2)}$ is fixed by

$$\delta_\mu = -\left[(N-1)\lambda + 3\lambda\right]\frac{\Gamma(1-d/2)}{(4\pi)} + \text{finite}$$

<u>in the limit of $d \to 2$</u>

$$\delta_\mu = -\lambda(N+2)\frac{\Gamma(1-d/2)}{(4\pi)} + \text{finite}$$

So let's now expand $\Gamma(-d/2) = \frac{\Gamma(2-d/2)}{d/2\,(d/2-1)}$ to investigate the pole at <u>$d=4$</u>

Now we have both the quadratic in ϕ_{cl} and also a quartic in ϕ_{cl}, which gives us the counter-terms

$$\delta_\mu = \left[2\cdot 3\lambda \cdot \frac{\Gamma(2-d/2)}{2(2-1)}\mu^2 + (N-1)(2\mu^2\lambda)\frac{\Gamma(2-d/2)}{2(2-1)}\right](4\pi)^2 + \text{finite}$$

|| In limit of $d=4$

$$\delta_\mu = \lambda\mu^2(N+2)\frac{\Gamma(2-d/2)}{(4\pi)^2} + \text{finite}$$

and for δ_λ we get

$$\delta_\lambda = 2\cdot\frac{\Gamma(2-d/2)}{2(2-1)}\frac{1}{(4\pi)^2}\left[(N-1)\lambda^2 + 9\lambda^2\right] = \lambda^2(N+8)\frac{\Gamma(2-d/2)}{(4\pi)^2} + \text{finite}$$

So in the limit of $d \to 4$, the counter terms are:

$$\boxed{\begin{aligned} \delta_m &= \lambda \mu^2 (N+2) \frac{\Gamma(2-d/2)}{(4\pi)^2} \\ \delta_\lambda &= \lambda^2 (N+8) \frac{\Gamma(2-d/2)}{(4\pi)^2} + \text{finite} \end{aligned}}$$

If we do this in $\overline{MS}$ - shceme (where counter-terms cancel $\underline{\frac{2}{\varepsilon} - \gamma + \ln(4\pi)}$)

$$\frac{\Gamma(-d/2)}{(4\pi)^{d/2}} (M^2)^{d/2} = \frac{1}{\frac{d}{2}(d/2 - 1)} \frac{\Gamma(2-d/2)}{(4\pi)^{d/2}} (M^2)^{d/2} \quad \Big\| \text{ expand in } d = 4-\varepsilon$$

$$= \frac{M^4}{2(4\pi)^2} \left(\frac{2}{\varepsilon} - \gamma + \ln(4\pi) - \ln(m^2) + \frac{3}{2} \right)$$

$\Rightarrow \delta_\mu \phi_{cl}^2 + \frac{1}{2}\delta_\lambda \phi_{cl}^4 - \frac{\Gamma(-d/2)}{(4\pi)^{d/2}} (M^2)^{d/2} \to \frac{M^4}{2(4\pi)^2} \left(-\ln\left(\frac{M^2}{\Lambda^2}\right) + 3/2 \right)$, where we take the counter terms to cancel the $\frac{1}{\varepsilon}$ divergence along with some constants, here Λ^2 is some arbitrary mass scale Parameter to make eq. dimensionally correct.

This makes it so that our potential is

$$V_{eff} = -\mathcal{L}_R[\phi_{cl}] - \frac{1}{4} \frac{1}{(4\pi)^2} \sum_i (M_i)^4 \left(-\ln\left(\frac{M_i^2}{\Lambda^2}\right) + 3/2 \right)$$

$$\begin{aligned} V_{eff} = -\mathcal{L}_R[\phi_{cl}] - \frac{1}{4}\frac{1}{(4\pi)^2} \Big[&-(N-1)(\mu^2 + \lambda\phi_{cl}^2)^2 \left(\ln\left(\frac{(\lambda\phi_{cl}^2 + \mu^2)}{\Lambda^2} \right) + 3/2 \right) \\ &- (\mu^2 + 3\lambda\phi_{cl}^2)^2 \left(\ln\left(\frac{\mu^2 + 3\lambda\phi_{cl}^2}{\Lambda^2} \right) + 3/2 \right) \Big] \end{aligned} \quad \Big\| \ \mathcal{L}_R[\phi_{cl}] = -\frac{1}{2}\mu^2\phi_{cl}^2 - \frac{\lambda}{4}\phi_{cl}^4$$

$$\boxed{\begin{aligned} \Rightarrow V_{eff} = \frac{1}{2}\mu^2\phi_{cl}^2 + \frac{\lambda}{4}\phi_{cl}^4 + \frac{1}{4}\frac{1}{(4\pi)^2} \Big\{ &(N-1)(\mu^2 + \lambda\phi_{cl}^2)^2 \left(\ln\left(\frac{\mu^2 + \lambda\phi_{cl}^2}{\Lambda^2} \right) - 3/2 \right) \\ &+ (\mu^2 + 3\lambda\phi_{cl}^2)^2 \left(\ln\left(\frac{\mu^2 + 3\lambda\phi_{cl}^2}{\Lambda^2} \right) - 3/2 \right) \Big\} \end{aligned}}$$

This is the effective potential when using $\overline{MS}$ - scheme, we notice that the result is same as Peskin (11.79) with $\mu^2 \leftrightarrow -\mu^2$ and $M^2 \leftrightarrow \Lambda^2$ being the difference in notation

So in this exercise we have first gotten equation for V_{eff} that has counter terms not yet fixed (middle of last page), then we fixed the counter-terms such that the counter-terms cancelled all of $\Gamma(-d/2)$ and not just the $1/\varepsilon$ - part.
The we kinda used $\overline{MS}$-scheme to get the final eq. for V_{eff}, but this is not by using the original counter terms we derived but by using the ones that only cancel $2/\varepsilon - \gamma + \ln(4\pi)$ terms, for this we kinda took the result from $\overline{MS}$ from Peskin without detailed calculation

since it was just to show that the resulting V_{eff} looks same as Peskin one and that the exercise was kinda completed when δ_m and δ_λ were derived. Also the original δ_m and δ_λ we got weren't final since the + finite terms depend on the renormalization condition. And we got result V_{eff} for one such renormalization condition which was $\overline{MS}$-sheme. Now we can use this V_{eff} in $\overline{MS}$-bar scheme in next exercise.

Exercise 1.8

In $\overline{MS}$-sheme:
$$V_{eff} = \frac{1}{2}\mu^2\phi_{cl}^2 + \frac{\lambda}{24}\phi_{cl}^4 + \frac{1}{4(4\pi)^2}\left\{(N-1)(\mu^2+\lambda\phi_{cl}^2)^2\left[\ln\left(\frac{\mu^2+\lambda\phi_{cl}^2}{\Lambda^2}\right) - 3/2\right] + (\mu^2+3\lambda\phi_{cl}^2)^2\left[\ln\left(\frac{\mu^2+3\lambda\phi_{cl}^2}{\Lambda^2}\right) - 3/2\right]\right\}$$

Now we investigate this when $\mu^2 = 0$

$$\Rightarrow V_{eff} = \frac{\lambda}{4}\phi_{cl}^4 + \frac{1}{4(4\pi)^2}\left\{(N-1)\lambda^2\phi_{cl}^4\left[\ln\left(\frac{\lambda\phi_{cl}^2}{\Lambda^2}\right) - \frac{3}{2}\right] + 9\lambda^2\phi_{cl}^4\left[\ln\left(\frac{3\lambda\phi_{cl}^2}{\Lambda^2}\right) - 3/2\right]\right\}$$
$$= \frac{\lambda}{4}\phi_{cl}^4 + \frac{\lambda^2\phi_{cl}^4}{4(4\pi)^2}\left(\ln\left(\frac{\lambda\phi_{cl}^2}{\Lambda^2}\right) - 3/2\right)\left\{N-1+9\right\} + \frac{9\lambda^2\phi_{cl}^4}{4(4\pi)^2}\ln(3)$$

$$V_{eff} = \frac{1}{4}\phi_{cl}^4\left\{\lambda + \frac{\lambda^2}{(4\pi)^2}\left((N+8)\left[\ln\left(\frac{\lambda\phi_{cl}^2}{\Lambda^2}\right) - 3/2\right] + 9\ln 3\right)\right\}$$

Now let's find the minimum with $\frac{\partial V_{eff}}{\partial\phi_{cl}} = 0$

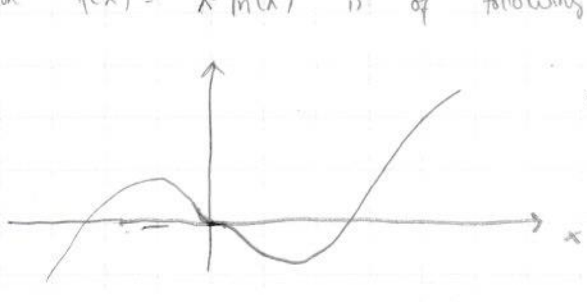
$$\frac{\partial V_{eff}}{\partial\phi_{cl}} = \phi_{cl}^3\left(\lambda + \frac{\lambda^2}{(4\pi)^2}\left((N+8)\left[\ln\left(\frac{\lambda\phi_{cl}^2}{\Lambda^2}\right) - 3/2\right] + 9\ln(3)\right)\right) + \frac{N+8}{4}\phi_{cl}^4\, 2\,\frac{1}{\phi_{cl}} = 0$$

$$= \phi_{cl}^3\left[\lambda + (N+8)/2 + \frac{\lambda^2}{(4\pi)^2}\left[9\ln(3) + (N+8)\left(\ln\left(\frac{\lambda\phi_{cl}^2}{\Lambda^2}\right) - 3/2\right)\right]\right] = 0$$

$$\frac{\partial V_{eff}}{\partial\phi_{cl}} = \phi_{cl}^3\left[\lambda + (N+8)/2 + \frac{\lambda^2}{(4\pi)^2}\left(9\ln 3 + (N+8)\left(2\ln(|\phi_{cl}|/\Lambda) + \ln\lambda - 3/2\right)\right)\right] = 0$$

as $\phi_{cl} \to 0$ $\ln(|\phi_{cl}|) \to -1/|\phi_{cl}|$ so $\frac{\partial V_{eff}}{\partial\phi_{cl}}$ does have extremum at $\phi_{cl} = 0$

but that is a saddle point and not the minimum. Since a graph for function $f(x) = x^3\ln(x^2)$ is of following form:



So at $x = 0$ we have a saddle point and the minima is at $x > 0$

We get the minimum or maximum from demanding the terms inside the bracket in $\frac{\partial V_{eff}}{\partial \phi_{cl}} = 0$ be 0 :

$$\lambda + (N+8)/2 + \frac{\lambda^2}{(4\pi)^2}\left(9\ln(3) + (N+8)\left[2\ln(|\phi_{cl}|/\Lambda) + \ln\lambda - 3/2\right]\right) = 0$$

From looking at the graph drawn earlier we know that the minimum is at $\phi_{cl} > 0$ so let's look only for $\phi_{cl} > 0$ case

$$\Rightarrow -\frac{\lambda + (N+8)/2}{\lambda^2/(4\pi)^2} - 9\ln(3) = N+8\left(2\ln\left(\frac{\phi_{cl}\lambda^{1/2}}{\Lambda}\right) - 3/2\right)$$

$$\Rightarrow \ln(\phi_{cl}\lambda^{1/2}/\Lambda) = \frac{1}{2}\left[\frac{3}{2} - \frac{1}{N+8}\left(9\ln(3) + \frac{2\lambda + (N+8)}{2\lambda^2/(4\pi)^2}\right)\right]$$

$$\Rightarrow \frac{\phi_{cl}\lambda^{1/2}}{\Lambda} = \exp\left\{\frac{1}{2}\left(\frac{3}{2} - \frac{9\ln 3}{N+8} - \frac{(4\pi)^2}{(N+8)\lambda} - \frac{(4\pi)^2}{2\lambda^2}\right)\right\}$$

$$\boxed{\Rightarrow \phi_{cl} = \Lambda\lambda^{-1/2}\exp\left\{\frac{1}{2}\left(\frac{3}{2} - \frac{9\ln 3}{N+8} - \frac{(4\pi)^2}{(N+8)\lambda} - \frac{(4\pi)^2}{2\lambda^2}\right)\right\}}$$

This is the minimum of V_{eff} and it is not at $\phi_{cl} = 0$

$\Rightarrow$ Symmetry is spontaneously broken, (assuming the perturbation theory holds

And the perturbation theory holds for only a small

$\left|\frac{\lambda(N+8)}{(4\pi)^2}\ln\left(\phi_{cl}^2/\Lambda^2\right)\right|$ since the potential can be written as

$$V_{eff} = \frac{\lambda}{4}\phi_{cl}^4\left(1 + \frac{\lambda}{(4\pi)^2}\left((N+8)\left[\ln\left(\frac{\lambda\phi_{cl}^2}{\Lambda^2}\right) - 3/2\right] + 9\ln(3)\right)\right)$$

$$V_{eff} = \frac{\lambda}{4}\phi_{cl}^4\left(1 + \underbrace{\frac{\lambda}{(4\pi)^2}(N+8)\ln\left(\frac{\phi_{cl}^2}{\Lambda^2}\right)}_{\text{Absolute value of this needs to be small for the perturbation theory to hold}} + \frac{\lambda}{(4\pi)^2}(N+8)\left[\ln(\lambda) - 3/2\right] + 9\ln(3)\right)$$